



Node.js

EXPRESS / FASTIFY ENGINE

MongoDB

MONGOOSE ODM + INDEXES

10:25 AM

ASIA/DHAKA CRON ENGINE

Socket.io

REAL-TIME EVENT STREAM

3-Tier Layered Backend Architecture

1. GATEWAY & ROUTING

Security & Auth Pipeline

- Helmet + CORS + Rate Limiter
- **JWT Bearer Verification & RBAC**

REST API Controllers

/api/v1/attendance/\*  
/api/v1/penalties/\*  
/api/v1/constitution/\*  
/api/v1/employees/\*  
/api/v1/disputes/\*

Strict Zod / Joi Request Validation

WebSocket / SSE Server

- Real-time delta broadcasting
- 10:25 AM prosecution alert
- Instant dispute status sync

**Socket.io Engine**

2. CORE DOMAIN SERVICES

Rules Engine (engine.js)

- Evaluates Present vs. WhatsApp Done
- **Auto-applies fine under Constitution**
- Idempotent bulk MongoDB upsert
- Asia/Dhaka 10:25 AM node-cron

Mapping & Fuzzy Matcher

- Resolves WA phone/nick → Emp ID
- Fallback tokenized fuzzy similarity

Constitution Service

- Manages Articles, Rules & Bylaws
- **Server-side PDF generation pipeline**

HRM & WhatsApp Connectors

- Token cache, auto-refresh, retry backoff

3. MONGODB COLLECTIONS

penalties

employeeId, date, amount, status,  
**Index: { date: 1, employeeId: 1, status: 1 }**

attendance\_logs

date, presentUsers[], doneUsers[]  
**Index: { date: -1 }**

constitution\_rules

articleNo, category, fineBDT, severity  
**Index: { articleNo: 1, category: 1 }**

employees & departments

officialName, waName, designationId

disputes & audit\_logs

penaltyId, reason, reviewerId, resolution

⚡ High Throughput

Express/Fastify + MongoDB connection pooling.  
Sub-10ms query latency on indexed lookups.

🔒 Enterprise RBAC

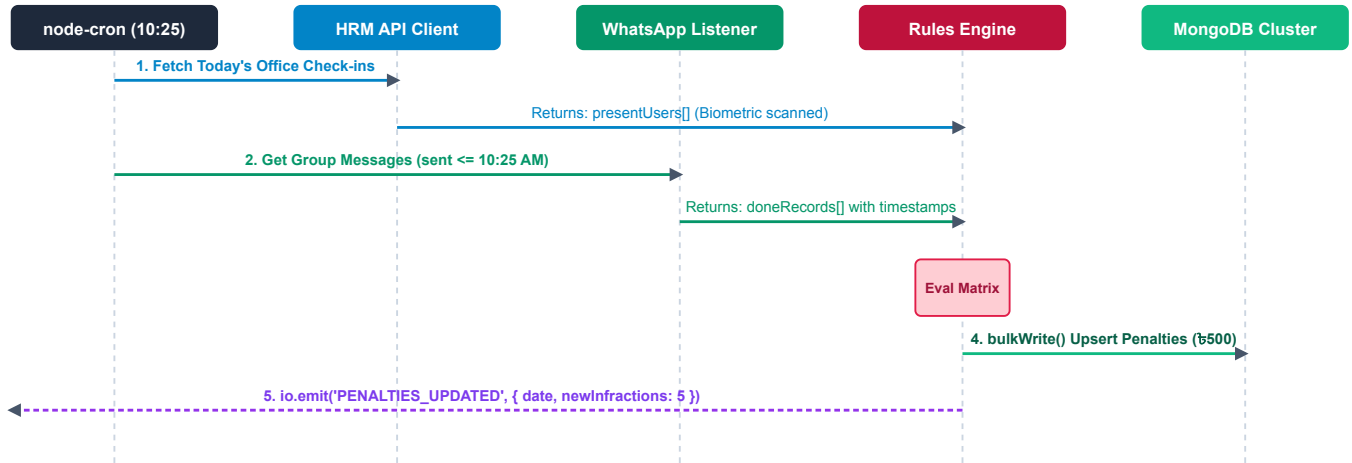
JWT authentication with granular module permissions: [view, create, update, delete] enforced at route level.

🔄 Idempotent Cron

10:25 AM job can run multiple times safely using MongoDB bulkWrite with upsert keys.



### 10:25 AM Distributed Execution Sequence



### Node.js Prosecution Service Implementation (`rules-engine/engine.js`)

```
async function evaluateDailyCompliance({ date, presentUsers, doneRecords }) {
  // 1. Filter WA messages sent on or before 10:25 AM
  const validDoneEmployees = new Set(doneRecords.filter(r => r.time <= '10:25:00').map(r => r.empId));

  // 2. Identify Non-Compliant: Present in office but no done in group
  const infractions = presentUsers.filter(u => !validDoneEmployees.has(u.employeeId));

  // 3. Idempotent bulk MongoDB upsert
  await PenaltyModel.bulkWrite(infractions.map(u => ({
    updateOne: {
      filter: { date, employeeId: u.employeeId, article: 'Art-1.1' },
      update: { $setOnInsert: { amount: 500, status: 'PENDING', createdAt: new Date() } },
      upsert: true
    }
  })));
}
```



## Collection: penalties

```
{
  _id: ObjectId,
  employeeId: String, // Ref: employees.employeeId
  articleNumber: String, // "Art-1.1"
  reason: String, // "Present but no done message"
  amount: Number, // 500 (BDT)
  status: String, // "PENDING" | "PAID" | "DISPUTED"
  date: String, // "YYYY-MM-DD"
  paidVia: String, // "CASH" | "PAYROLL" | null
  createdAt: Date, updatedAt: Date
}
INDEX: { date: 1, employeeId: 1, status: 1 }
```

## Collection: constitution\_rules

```
{
  _id: ObjectId,
  articleNumber: String, // "Art-1.1" (Unique)
  category: String, // "Attendance" | "Security"
  title: String, // "Morning Standup Check-in"
  description: String,
  fineAmountBDT: Number, // 500
  severity: String, // "STANDARD" | "SEVERE"
  isAutomated: Boolean, // true
  updatedAt: Date
}
INDEX: { articleNumber: 1, category: 1 }
```

## Collection: attendance\_logs

```
{
  _id: ObjectId,
  date: String, // "YYYY-MM-DD" (Unique)
  presentCount: Number,
  doneCount: Number,
  penaltyCount: Number,
  records: [{
    employeeId: String, present: Boolean,
    done: Boolean, penalty: Boolean, timeIso: String
  }]
}
INDEX: { date: -1 }
```

## Collection: employees & users

```
{
  _id: ObjectId,
  employeeId: String, // "E-104" (Unique)
  officialName: String, email: String,
  whatsappName: String, phone: String,
  departmentId: ObjectId, designationId: ObjectId,
  role: String, // "ADMIN" | "MANAGER" | "VIEWER"
  permissions: Object // Granular CRUD matrix
}
INDEX: { employeeId: 1, whatsappName: 1 }
```

## High-Speed MongoDB Dashboard Aggregation Pipeline





Core Backend REST Endpoints (Communicating with Frontend)

| Method | Endpoint URI                    | Payload / Query Params                         | Response Description                                          |
|--------|---------------------------------|------------------------------------------------|---------------------------------------------------------------|
| GET    | /api/v1/attendance/today        | None (Bearer Token)                            | Returns today's biometric presence + WhatsApp status          |
| GET    | /api/v1/penalties               | ?status=PENDING&dateStart=...&page=1           | Paginated penalties grouped by employee with BDT totals       |
| POST   | /api/v1/penalties               | { employeeId, articleNumber, amount, date }    | Manual or system infraction creation                          |
| PATCH  | /api/v1/penalties/:id/status    | { status: "PAID"   "DISPUTED", note }          | Updates penalty lifecycle state + emits WebSocket event       |
| GET    | /api/v1/constitution            | ?category=Attendance                           | Full articles, bylaws, fine schedules, and severity codes     |
| POST   | /api/v1/integrations/hrm/export | { month: "2026-09", cutoffDate: "2026-09-25" } | Pushes verified penalties to HRM Payroll for salary deduction |



WhatsApp Daemon (whatsapp-web.js)

- Runs as headless background client with persistent auth state (.webjs\_auth).
- In-memory message ring buffer + message deduplication.
- Outbound template notifier for instant infraction dispatch.
- Auto-reconnect with QR scan terminal fallback.



HRM Attendance & Payroll Bridge

- Daily Ingestion: /api/brotecs/today-dashboard.
- JWT Token Cache: File + In-Memory cache with auto-refresh.
- Exponential retry backoff on corporate network fluctuations.
- Safe month-end payroll deduction reconciliation.